Supervised Machine Learning- Regression

Assumptions of Regression and Model Evaluation

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Agenda

- Assumptions of Linear Regression
- Tests for Assumptions of Linear Regression
- Model Evaluation Metrics
- Presence of Categorical Variables
- Interaction Effect

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Assumptions of Linear Regression

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Assumptions of linear regression

- The dependent variable must be numeric
- Linear relationship between dependent and independent variables
- Predictors must not show multicollinearity
- Independence of observations should exist (Absence of Autocorrelation)
- The error terms should be homoscedastic
- The error terms must follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Assumptions of linear regression

- Numeric dependent variable
- Absence of Multicollinearity

Assumption to be checked before building a model

Build a model

- Linear relationship
- Absence of Autocorrelation
- Error terms should be homoscedastic
- Error terms must follow $N(0,\sigma^2)$

Assumption to be checked after building a model

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests for Assumptions of Linear Regression

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests before model building

- The dependent variable must be numeric

- Predictors must not show multicollinearity

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Is the dependent variable numeric?

- Regression Analysis requires the target variable to be numeric in nature
- For example: returns, sales of a product, yield of a crop, risk in financial services
- In context with our example, we see that Premium is numeric

Mileage	Premium (in dollars)
15	392.5
14	46.2
17	15.7
7	422.2
10	119.4
7	170.9
20	56.9
21	77.5
18	214
11	65.3
7.9	250
8.6	220
12.3	217.5
17.1	140.88
19.4	97.25

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q & A

Question:

If our target variable is: 0, 1, 1, 0, 0, where 0 indicates presence of a disease and 1 indicates absence.

Is it appropriate to use regression to find the whether the person has a disease?

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Q & A

Question:
If our target variable is: 0, 1, 1, 0, 0, where 0 indicates presence of a disease and 1 indicates absence. Is it appropriate to use regression to find the whether the person has a disease?

Answer:
No. Because the target variable is a categorical variable. Thus, it is a classification problem.

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests before model building

- The dependent variable must be numeric

- Predictors must not show multicollinearity

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

What is multicollinearity?

- Multicollinearity arises when the independent variables have high correlation among each other

- Multicollinearity may be introduced if there exists empirical relationship among variables such as income = expenditure + saving

- In presence of it, the best fit line obtained from OLS method is no more “best”

- Also, the confidence interval obtained for β’s is wider since the SE(β) becomes large

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Multicollinearity detection

- Determinant of correlation mat
- Condition Number (CN)

Is there multicollinearity present?

- Correlation matrix
- Variance Inflation Number (VIF

Which variables are involved in multicollinearity?

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Is there multicollinearity?

Determinant of the correlation matrix:

Let D be the determinant of correlation matrix. Then $0 < D < 1$

D=0	High multicollinearity
D=1	No multicollinearity

Condition Number (CN):

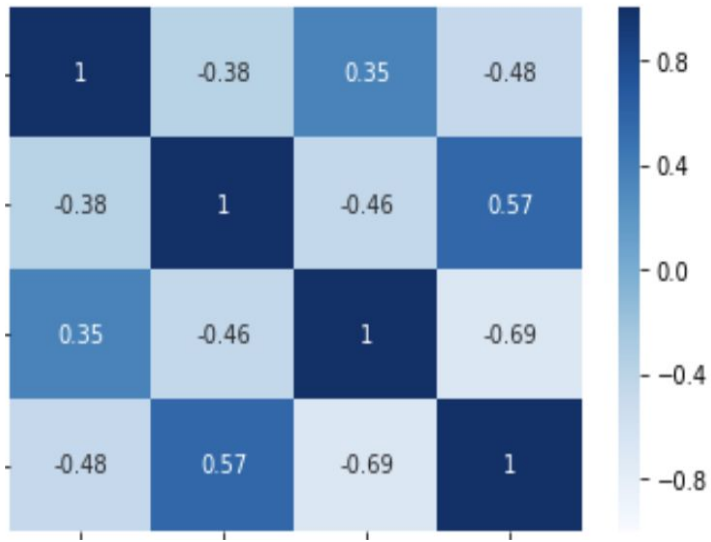
$CN > 1000$	Severe multicollinearity
$100 < CN < 1000$	Moderate multicollinearity
$100 < CN$	No multicollinearity

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Which variables are involved in multicollinearity?

Correlation matrix:

If the off-diagonal values tend to ±1 then it indicates high correlation between the variable pair. However this inspection is not enough



This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Which variables are involved in multicollinearity?

Variance Inflation Factor (VIF):

$$VIF = \frac{1}{1-R^2}$$

Where R^2 is obtained by regressing a predictor variable over all the other predictors in the model

Value	Interpretation
VIF > 5	High correlation
5 > VIF >1	Moderate correlation
VIF = 1	No correlation

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests after model building

- Linear relationship between dependent and independent variables
- Independence of observations should exist (i.e. Absence of Autocorrelation)
- The error terms should be homoscedastic
- The error terms must follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests after model building

- Linear relationship between dependent and independent variables
- Independence of observations should exist (i.e. Absence of Autocorrelation)
- The error terms should be homoscedastic
- The error terms must follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Assumption of linearity

- An assumption of linear regression is that it should be linear in the parameter

- The independent variables must have a linear relationship with the dependent variable

- The residuals and the fitted values should be independent

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Existence of linear relationship

An assumption of linear regression is that it should be linear in the parameter

Linear Relationship

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2 + \varepsilon$

$y = \beta_0 - \beta_1 \log(x_1) + \beta_2 x_2 + \varepsilon$

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 - \beta_3 x_1 x_2 + \varepsilon$

Nonlinear Relationship

$y = \beta_0 - e^{\beta 1 x 1} + \varepsilon$

$y = \beta_0 x_1 / \beta_1 x_1 + \varepsilon$

$y = \beta_0 + x_1^{\beta 1} . x_2^{\beta 2} + \varepsilon$

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Existence of linear relationship

- The independent variables must have a linear relationship with the dependent variable

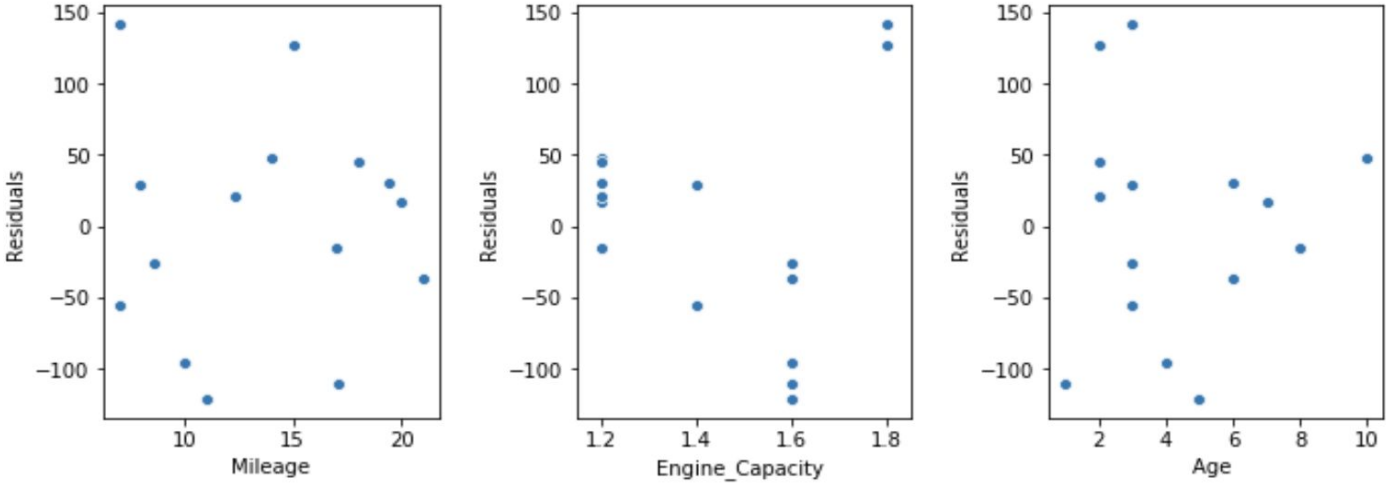
- This can be checked plotting a scatter plot of residuals vs predictors

- A scatter plot depicting no pattern indicates that the variable has a linear relationship with the response variable

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Existence of linear relationship

In context with our data, we see a random pattern in all the three plots. Hence, we may say that, the predictors are linearly related with the response variable.

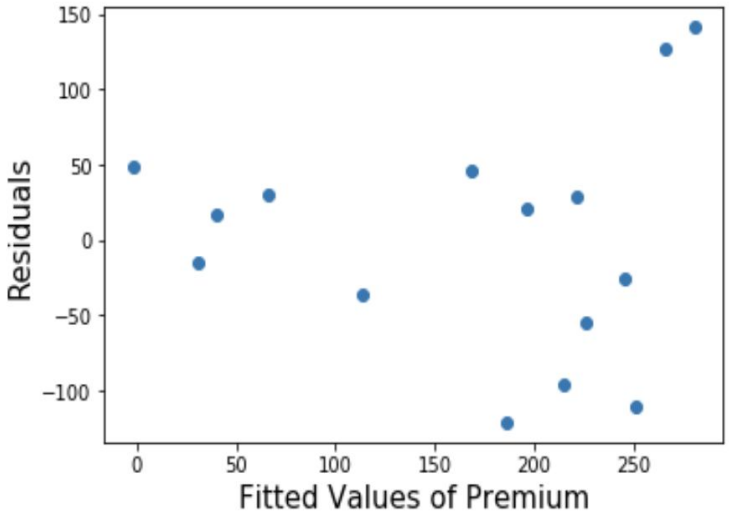


This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Existence of linear relationship

- The plot of residuals against the fitted tells the presence of linear relationship

- For linear relationship, the points must be at random, i.e., it should not exhibit much distinctive pattern, no non-linear trends or changes in variability



This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests after model building

- Linear relationship between dependent and independent variables
- Independence of observations should exist (i.e. Absence of Autocorrelation)
- The error terms should be homoscedastic
- The error terms must follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Assumption of autocorrelation

{Auto}correlation

self

- Assumption of autocorrelation is violated when residuals are correlated within themselves, i.e. they are serially correlated
- Autocorrelation does not impact the regression coefficients but the associated standard errors are reduced
- This reduction in standard error leads to a reduction in associated p-value
- It incorrectly concludes that a predictor is statistically significant

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Causes of autocorrelation

- Some important variables are not considered in the data
- If the relationship between the target and predictor variables is non-linear and is incorrectly considered linear
- Presence of carry over effect

  Example: The additional expenses from the budget for last month are carried over in creating the budget for next month

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Durbin - Watson Test

- To test whether the error terms are autocorrelated,we Durbin-Watson test
- We test whether autocorrelation is present or not
- The hypothesis is given by:

H_0:The error terms are not autocorrelated

against

H_1:The error terms are autocorrelated

- Failing to reject H_0, will imply that the error terms are autocorrelated

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Durbin - Watson test

The test statistic is given by

Residual of tth observation

Residual of $t\text{-}1^{th}$ observation

$$d = \frac{\sum \hat{e}_t - \hat{e}_{t-1}}{\sum \hat{e}_t^2}$$

$d \in [0,4]$

Value	Interpretation
0 < d <2	Positive autocorrelation
d = 2	No autocorrelation
2 < d < 4	Negative autocorrelation

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests after model building

- Linear relationship between dependent and independent variables
- Independence of observations should exist (i.e. Absence of Autocorrelation)
- The error terms should be homoscedastic
- The error terms must follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

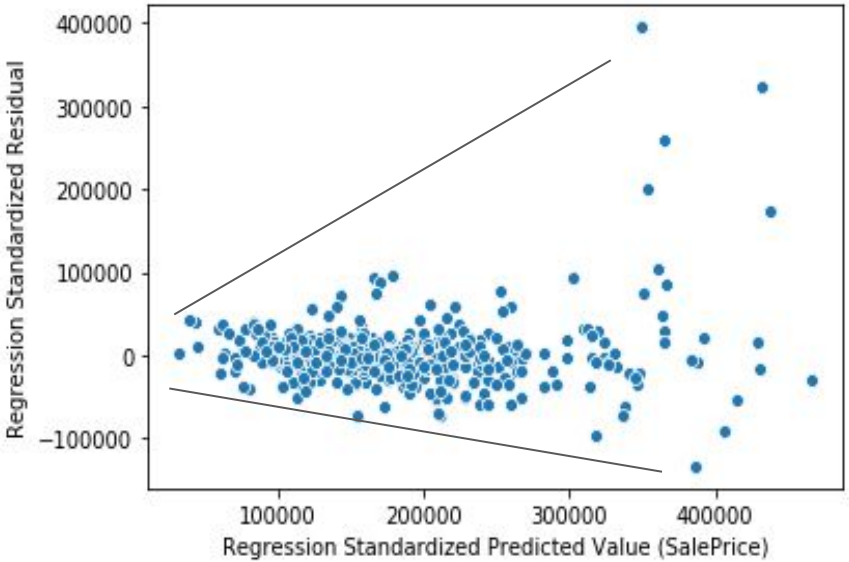
Homoscedasticity assumption

$$\underbrace{\text{Homo}}_{\text{Same}}\underbrace{\text{scedasticity}}_{\text{Variance}}$$

- Variance of the residual is assumed to be independent of the explanatory variables
- Heteroscedasticity: non-constant variance of residuals
- It happens due to the presence of extreme values

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

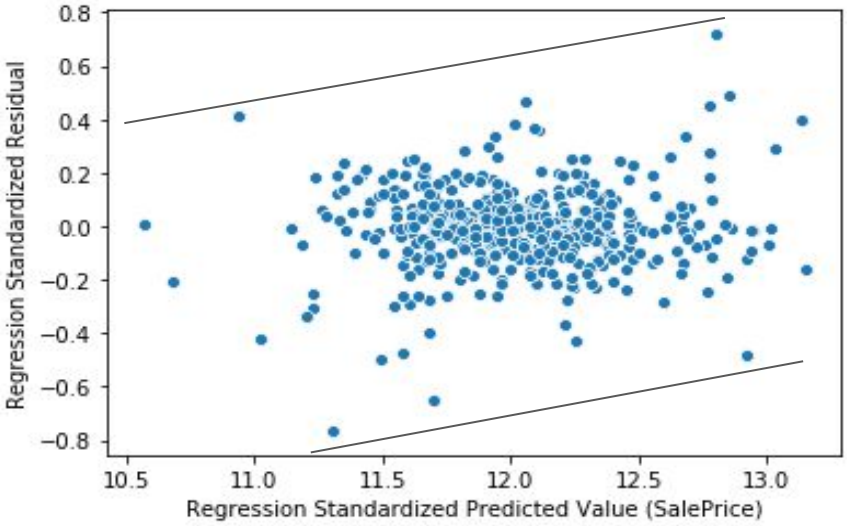
Heteroscedasticity



- Funnel type shape is seen in the graph
- Hence we can say that there is a presence of “Heteroscedasticity”

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Homoscedasticity



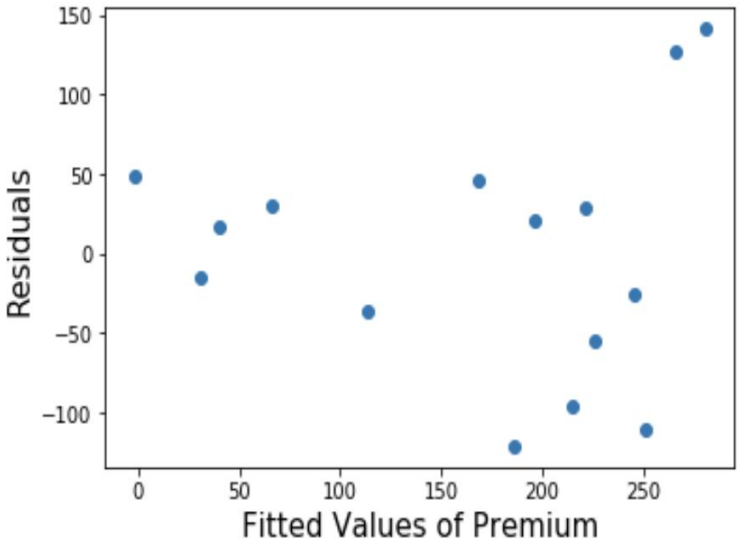
- There is no visible funnel or bow type pattern in the plot
- We can see presence of “Homoscedasticity”

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Residual vs Fitted plot

- The plot of residuals against the fitted values tells whether the error terms have equal variance

- It should look random, i.e., it should not exhibit much distinctive pattern, no non-linear trends or changes in variability



This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Homoscedasticity

The statistical test to test for the homoskedasticity of the errors are

- Goldfeld Quandt test
- Breusch Pagan test

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Goldfeld-Quandt test

- For presence of a constant variance of error terms, i.e. to test

H_0: The errors terms are homoskedastic against H_1: The errors terms are heteroskedastic

- Decision rule: Reject H_0, if the p-value associated with test statistic is less than α (level of significance), which implies that there is heteroskedastic, i.e. the error terms have do not equal variance

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Breusch Pagan test

- For presence of a constant variance of error terms, i.e. to test

H_0: The errors terms are homoskedastic

against

H_1: The errors terms are heteroskedastic

- Decision rule: Reject H_0, if the p-value associated with test statistic is less than α (level of significance), which implies that there is heteroskedastic, i.e. the error terms have do not equal variance

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Tests after model building

- Linear relationship between dependent and independent variables
- Independence of observations should exist (i.e. Absence of Autocorrelation)
- The error terms should be homoscedastic
- The error terms must follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Normality test

- Parametric statistical methods assume that the underlying data has a normal distribution

- Normality tests are used to determine if a data set is well-modeled by a normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Normality testing techniques

- Quantile-Quantile Plot
- Jarque-Bera (JB) Test
- Shapiro-Wilk Test

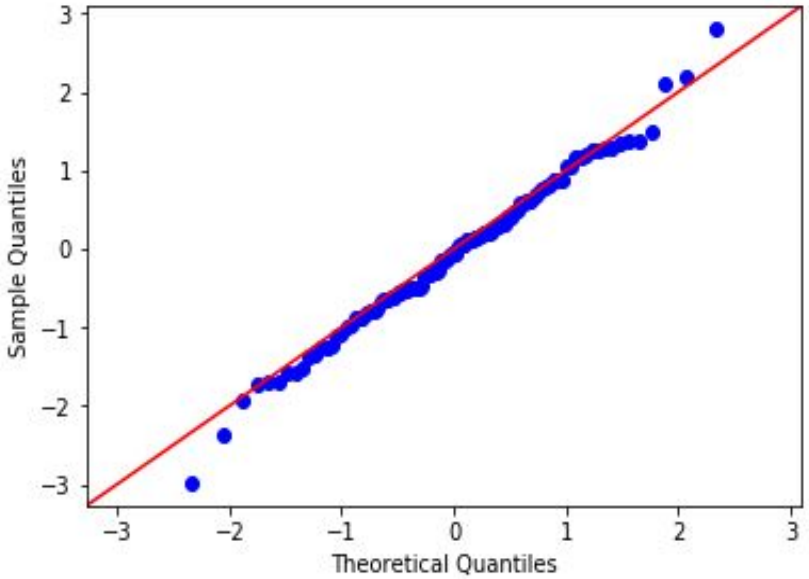
This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Quantile-Quantile Plot (QQ plot)

- Used to determine whether two datasets follow the same distribution
- The quantiles of two datasets are plotted against each other
- A reference line is plotted at 45^0
- If the points lie on the reference line we conclude they follow the same distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

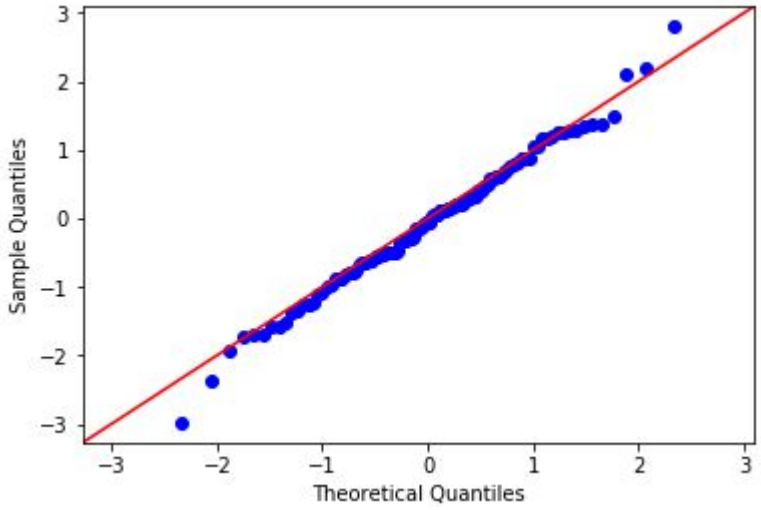
Normal QQ plot



- The x axis has points from a theoretically calculated normal distribution
- They are compared with sample data on the y axis
- If the sample data has a normal distribution the points lie on the reference line

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Normal QQ plot



- The x axis has points from a theoretically calculated normal distribution
- They are compared with sample data on the y axis
- If the sample data has a normal distribution the points lie on the reference line

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

JB test

- To test whether the data follows normal distribution, we test whether the skewness and kurtosis of the data are same as that of the normal distribution, i.e. to test

H_0: Skewness (S) = 0 and Kurtosis (K) = 0

against

H_1: Skewness (S) ≠ 0 and Kurtosis (K) ≠ 0

- Failing to reject H_0, implies that the data does not follow normal distribution

$$JB = \frac{n}{6}\left(S^2 + \frac{1}{4}(K-3)^2\right)$$

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

JB test

- The test statistics for n observations is given by

$$JB = \frac{n}{6}\left(S^2 + \frac{1}{4}(K-3)^2\right)$$

Kurtosis coefficient

Sample skewness

- The test statistics asymptotically follows chi squared distribution with 2 degrees of freedom (χ^2_2)

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Shapiro-Wilk test

- To test whether the data follows normal distribution, i.e. to test

H_0: The data is normally distributed

against

H_1: The data is not normally distributed

- Failing to reject H_0, implies that the data does not follow normal distribution

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Shapiro-Wilk test statistic

The test statistic is given by

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_i\right)^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2}$$

- n = sample size
- a_i = values computed from n samples (of size n each) from normal distribution based on their means, covariance matrix
- x_i = i^{th} ordered sample values
- $\bar{x}$ =sample mean

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Model Evaluation Metrics

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Model evaluation metrics

The model evaluation metrics are

- R^2
- Adjusted R^2
- The F test for overall significance

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

R-squared

- The R^2 value gives the percentage of variation in the response variable explained by the predictor variables

- If the values of R^2 = 0.87, it implies that 87% of variation in the response variable is explained by the predictor variables

$$R^2 = \frac{\text{Explained variation}}{\text{Total variation}} = \frac{\text{SSR}}{\text{SST}}$$

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Adjusted R-squared

- Adjusted R^2 gives the percentage of variation explained by independent variables that actually affect the dependent variable

- If the values of R^2 = 0.87, it implies that 87% of variation in the response variable is explained by the predictor variables

$$R^2_{adj} = 1 - \frac{(1-R^2)(n-1)}{n-k-1}$$

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

F test

- To check the significance of the regression model we use the F test
- It is similar to ANOVA for regression
- The test hypothesis is given by

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \quad = \beta_n = 0$$

against

$$H_1 : \beta_i > 0 \text{ or } \beta_i < 0$$

for at least one of the i values

- Failing to reject H_0, implies that the model is not significant

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

F statistic

- The test statistics is given by

$$Fstat = \frac{\text{(SST-SSE)/k}}{\text{SSE/(n-k-1)}}$$

$$n = \text{sample size}$$

$$k = \text{number of predictor variables}$$

- Decision rule: Reject H_0, if $F_0 > F_{(k,n-k-1),\alpha}$ or if the p-value is less than the α (level of significance)

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Presence of categorical variable

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Linear regression of categorical variable

- The regression method fails in presence of categorical variable

- Thus we need to convert the categorical variable to numeric variable

- In order to so, we use N - 1 dummy encoding

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

N-1 dummy encoding

- Dummy variables are binary variables used to represent categorical data
- For a categorical variable that can take k values k-1 dummy variables need to be created
- Dummy variable is assigned 1 if it takes a particular value else it is assigned 0

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Dummy variable example

Consider a variable, Gender, used to represent the gender of a citizen during the census

Gender: Male, Female

Since Gender takes 2 values it can be represented with 1 dummy variable D_1 as:

Value	D_1
Male	0
Female	1

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Data

Let us consider a categorical variable Manufacturer in the data and find out how it behaves.

Mileage	Manufacturer	Premium (in dollars)
15	Ford	392.5
14	Honda	46.2
17	Tata	15.7
7	Ford	422.2
10	Ford	119.4
7	Tata	170.9
20	Tata	56.9
21	Honda	77.5
18	Honda	214
11	Tata	65.3
7.9	Ford	250
8.6	Tata	220
12.3	Tata	217.5
17.1	Ford	140.88
19.4	Honda	97.25

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Example

- In context with our example, the categorical variable Manufacturer takes values Ford, Honda and Tata

- Since Manufacturer takes 3 values, two dummy variables Mfr_Honda and Mfr_Tata are created

Value	Mfr_Honda	Mfr_Tata
Ford	0	0
Honda	1	0
Tata	0	1

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Model with categorical variable

Now our model is

$$\text{Premium} = \beta_0 + \beta_1 \text{ Mileage} + \beta_2 \text{ Mfr_Honda} + \beta_3 \text{ Mfr_Tata} + \varepsilon$$

Parameter	Description
β_0	Premium value where the best fit line cuts the Y-axis (Premium)
β_1	Regression coefficient of the variable Mileage
β_2	Regression coefficient of the dummy variable Mfr_Honda
β_3	Regression coefficient of the dummy variable Mfr_Tata

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Linear regression model (dummy variable)

Mileage	Manufacturer	Premium (in dollars)
15	Ford	392.5
14	Honda	46.2
17	Tata	15.7
7	Ford	422.2
10	Ford	119.4
7	Tata	170.9
20	Tata	56.9
21	Honda	77.5
18	Honda	214
11	Tata	65.3
7.9	Ford	250
8.6	Tata	220
12.3	Tata	217.5
17.1	Ford	140.88
19.4	Honda	97.25

Based on the data, the β parameters are:

$\beta_0 = 368.93$, $\beta_1 = -9.117$,

$\beta_2 = -95.174$ and $\beta_3 = -129.216$

Thus the model is

$$Y = 368.93 - 9.117\, x_1 - 95.174\, x_2 - 129.216\, x_3$$

That is,

Premium = 368.93 - 9.117 Mileage - 95.174 Mfr_Honda - 129.216 Mfr_Tata

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Regression line (dummy variable)

The regression line:

$$\text{Premium} = \beta_0 + \beta_1 \text{ Mileage} + \beta_2 \text{ Mfr_Honda} + \beta_3 \text{ Mfr_Tata} + \varepsilon$$

If the manufacturer is Honda, the regression line becomes:

$$\begin{aligned}\text{Premium} &= \beta_0 + \beta_1 \text{ Mileage} + \beta_2 \text{ Mfr_Honda} + \beta_3 \text{ Mfr_Tata} \\ &= \beta_0 + \beta_1 \text{ Mileage} + \beta_2 (1) + \beta_3 (0) \\ &= \beta_0 + \beta_1 \text{ Mileage} + \beta_2 + 0 \\ &= (\beta_0 + \beta_2) + \beta_1 \text{ Mileage}\end{aligned}$$

Value	Mfr_Honda	Mfr_Tata
Ford	0	0
Honda	1	0
Tata	0	1

Note the change in the intercept value.

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Regression line (dummy variable)

The regression line:

$$\text{Premium} = \beta_0 + \beta_1 \text{ Mileage} + \beta_2 \text{ Mfr_Honda} + \beta_3 \text{ Mfr_Tata} + \varepsilon$$

Value	Mfr_Honda	Mfr_Tata
Ford	0	0
Honda	1	0
Tata	0	1

For manufacturer = Ford,

$\text{Premium} = \beta_0 + \beta_1 \text{ Mileage}$

Actual intercept

For manufacturer = Honda,

$\text{Premium} = (\beta_0 + \beta_2) + \beta_1 \text{ Mileage}$

Change in intercept

For manufacturer = Tata,

$\text{Premium} = (\beta_0 + \beta_3) + \beta_1 \text{ Mileage}$

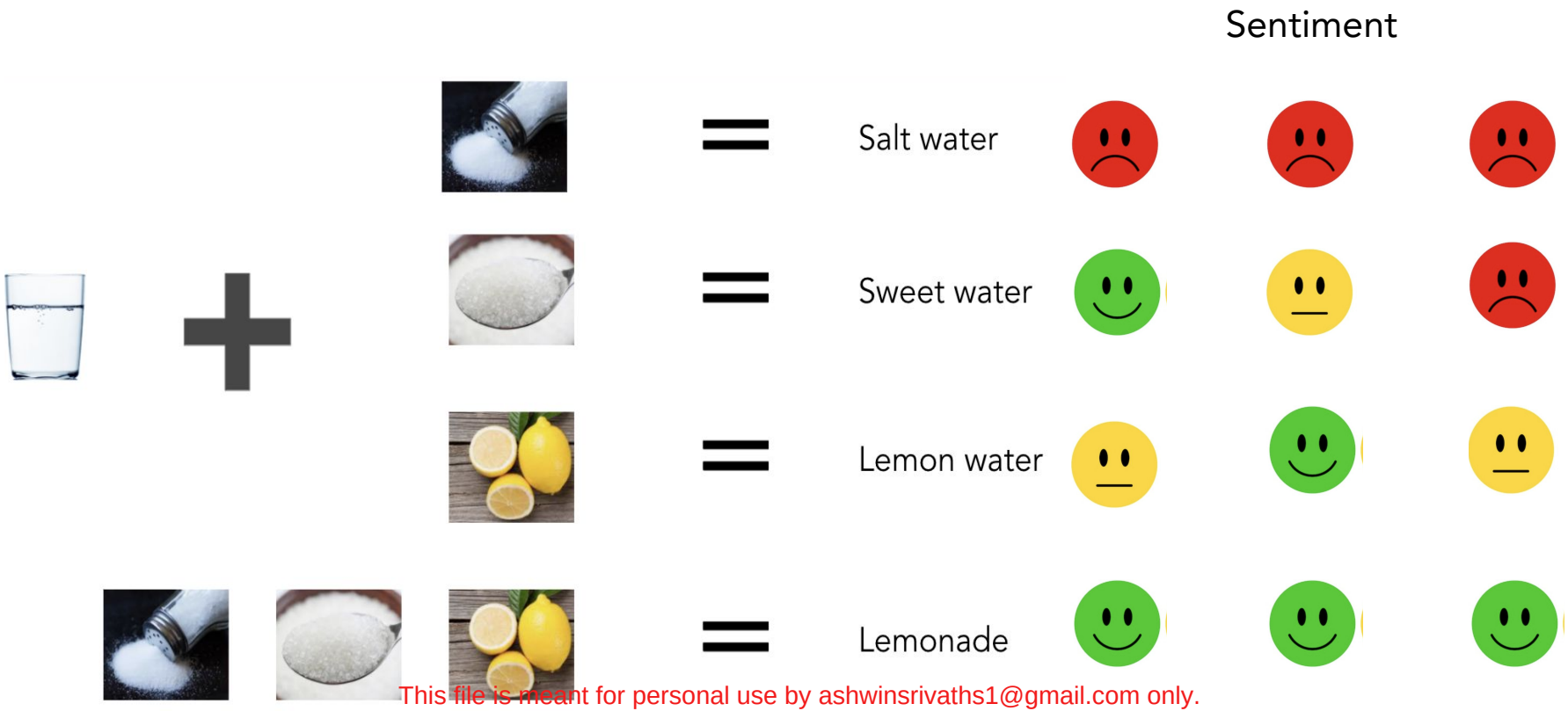
Change in intercept

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Interaction Effect

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Interaction effect



This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Interaction

- An interaction effect occurs when the effect of one variable depends on another variable. This combined effect may or may not improve the performance of the model

- Note: It does not imply that the predictor variables are collinear

Example: Salary of an employee increases with experience, but this may vary based whether the person has completed additional courses like MBA

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Interaction Effect

- In context with our example, we shall consider the interaction effect of variables Engine_Capacity and Mileage

- We obtained Int_EC_Mil by taking the product of Mileage and Engine_Capacity

- Let us check whether the interaction term is adding value to our model

Mileage	Engine_Capacity	Int_EC_Mil	Age	Premium (in dollars)
15	1.8	27	2	392.5
14	1.2	16.8	10	46.2
17	1.2	20.4	8	15.7
7	1.8	12.6	3	422.2
10	1.6	16	4	119.4
7	1.4	9.8	3	170.9
20	1.2	24	7	56.9
21	1.6	33.6	6	77.5
18	1.2	21.6	2	214
11	1.6	17.6	5	65.3
7.9	1.4	11.06	3	250
8.6	1.6	13.76	3	220
12.3	1.2	14.76	2	217.5
17.1	1.6	27.36	1	140.88
19.4	1.2	23.28	6	97.25

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Interaction Effect

Now our model is

$$\text{Premium} = \beta_0 + \beta_1 \text{ Mileage} + \beta_2 \text{ Engine_Capacity} + \beta_3 \text{Age} + \beta_4 \text{ Int_EC_Mil} + \varepsilon$$

Parameter	Description
β_0	Premium value where the best fit line cuts the Y-axis (Premium)
β_1	Regression coefficient of the variable Mileage
β_2	Regression coefficient of the variable Engine_Capacity
β_3	Regression coefficient of the variable Age
β_4	Regression coefficient of the variable Int_EC_Mil

Mileage	Engine_Capacity	Int_EC_Mil	Age	Premium (in dollars)
15	1.8	27	2	392.5
14	1.2	16.8	10	46.2
17	1.2	20.4	8	15.7
7	1.8	12.6	3	422.2
10	1.6	16	4	119.4
7	1.4	9.8	3	170.9
20	1.2	24	7	56.9
21	1.6	33.6	6	77.5
18	1.2	21.6	2	214
11	1.6	17.6	5	65.3
7.9	1.4	11.06	3	250
8.6	1.6	13.76	3	220
12.3	1.2	14.76	2	217.5
17.1	1.6	27.36	1	140.88
19.4	1.2	23.28	6	97.25

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Linear regression model (interaction effect)

Based on the data, the β parameters are:

$\beta_0 = -502.011$, $\beta_1 = 40.306$, $\beta_2 = 568.723$,

$\beta_3 = -25.781$ and $\beta_4 = -30.547$

Thus the model is

$$Y = 502.011 + 40.306\, x_1 + 68.723\, x_2 - 25.781 x_3 - 30.547\, x_4$$

Mileage	Engine_Capacity	Int_EC_Mil	Age	Premium (in dollars)
15	1.8	27	2	392.5
14	1.2	16.8	10	46.2
17	1.2	20.4	8	15.7
7	1.8	12.6	3	422.2
10	1.6	16	4	119.4
7	1.4	9.8	3	170.9
20	1.2	24	7	56.9
21	1.6	33.6	6	77.5
18	1.2	21.6	2	214
11	1.6	17.6	5	65.3
7.9	1.4	11.06	3	250
8.6	1.6	13.76	3	220
12.3	1.2	14.76	2	217.5
17.1	1.6	27.36	1	140.88
19.4	1.2	23.28	6	97.25

That is,

Premium = 502.011 + 40.306 Mileage + 68.723 Engine_Capacity - 30.54 Age - 25.781 Int_EC_Mil

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Thank You

This file is meant for personal use by ashwinsrivaths1@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.